The Geometric Atom: the Hydrogen State Space as a Label Lattice

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We study a discrete labelling of the hydrogen atom's bound-state space. The states $|n, \ell, m\rangle$ are indexed by a label lattice, drawn as a paraboloid (a visualization in which the depth coordinate $-1/n^2$ is placed by hand, not an output), on which the $SU(2) \times SU(1,1)$ ladder operators act as sparse matrices. This single-particle model separates two kinds of content: **(i) Algebraic exactness: [SYMBOLIC PROOF]** Transition operators $T_\pm, L_\pm$ reproduce the Rydberg quantum numbers exactly from operator eigenvalues (a two-line commutator identity, checked in exact arithmetic): the $SU(1,1)$ element $N = -2[T_+, T_-]$ returns n on the interior shells $n < n_{\max}$ (not on the truncated top shell), and L^2 returns $\ell(\ell+1)$ on every shell; on every shell, including the top one, the $SO(4)$ Casimir $\mathbf{L}^2 + \mathbf{A}^2 = n^2 - 1$ ($\mathbf{A}$ the energy-normalized Runge–Lenz vector) is exact on the labels. The energies $E_n = -1/(2n^2)$ are an imposed map from n (Fock's energy shell $p_0^2 = -2E$, with $p_0 = Z/n$ fixed by Fock's kernel eigenvalue, Paper 7), not an output of the ladder operators; once Fock's kernel fixes the function, on the labels they are $E_n = -1/(2(C+1))$ of the Casimir $C = n^2 - 1$ (i.e. $H = -\frac{1}{2}(1 - \Delta_{S^3})^{-1}$ at $Z = 1$). **(ii) Geometric artifacts: [MEASURED]** A node-amplitude proxy for the s/p gap is nonzero at finite lattice cutoff (37% at $n_{\max} = 5$, oscillating through 15.6% at 8 and 1.7% at 10 to 0.39% at 30 on the binary production lattice; see the §III QA note). **[INTERNAL THEOREM]** Its size tracks the residue of $n_{\max}$ modulo 3 — exactly closed form on $n_{\max} \equiv 0$, and generically larger off it (the off-/on-branch ratio grows asymptotically as $n_{\max}/(\sqrt{3}\,\pi)$; what was retracted is two earlier *precision* claims about that ratio [retracted 2026-09-04: sp-ratio-false-precision], read off three sampled cutoffs) — so the quoted endpoint sits on a favourable branch and the sequence is not evidence of convergence (2026-09-03; the earlier wording, "the graph Laplacian lifts s/p degeneracy ... through differential weighted connectivity", both contradicted the clause below and named the weighted construction's mechanism for binary-lattice numbers, where every edge weight is 1). The cutoff sweep from $n_{\max} = 5$ to 30 shows this is a discretization artifact—a residue-controlled truncation effect, *not* spectral aliasing [retracted 2026-09-04: sp-splitting-aliasing-mechanism] (that reading was withdrawn 2026-09-03: the l-blocks are disconnected and carry no degeneracy to alias)—that oscillates and reads 0.39% at $n_{\max} = 30$ on the $n_{\max} \equiv 0 \pmod 3$ branch (1.65% and 2.58% at 28 and 29): a proxy value, not evidence of recovering the exact Coulomb degeneracy [retracted 2026-09-04: sp-splitting-aliasing-mechanism], which belongs to the continuum operator (the graph's l-blocks are disconnected, so the graph has no s/p degeneracy of its own). **(iii) Geometric phase structure: [SYMBOLIC PROOF]** The Berry phase vanishes identically for the real-valued transition operators; on the optional topological-weight lattice ($w = 1/(n_1 n_2)$) the edge-weight log-holonomy $\Theta(n) = -2\ln((n+1)/n) \sim n^{-1}$ measures the variation (a discrete gradient) of that weight function (on the binary production lattice and the CG-magnitude construction $\Theta \equiv 0$). **This model describes only the electron's state space:** it contains no electromagnetic coupling, and we find no trace of the fine structure constant α in it (a companion paper [8] records a separate numerical Observation about α, not a coupling mechanism).

I. INTRODUCTION: THE TEXTURE OF THE VACUUM

Quantum mechanics is ordinarily formulated on an infinite-dimensional Hilbert space with continuous operators. The bound-state sector of hydrogen, however, is also completely described by a discrete set of labels (n, l, m) together with the operators that act on them; the continuous and discrete descriptions are dual descriptions of the same quantum-number structure, and we assign neither ontological priority.

We study a discrete labelling of the hydrogen state space. The hydrogen state space is indexed by a **graph**: nodes are the quantum-number labels of the basis states, edges connect labels related by the ladder operators $T_\pm, L_\pm$, and those operators act as sparse matrices on the labels.

This discrete structure can be drawn with a natural geometry. The hydrogen atom's $SO(4)$ symmetry [4], extended to the $SO(4,2)$ dynamical group [1–3], motivates the **paraboloid lattice** picture, in which the labels (n, l, m) are placed at coordinates on a curved 3D surface. The picture is a visualization, not a derivation: only the $SU(2) \times SU(1,1)$ subalgebra is used below, and the depth coordinate $z = -1/n^2$ is placed by hand from the known Rydberg formula (it is not an output of the lattice).

The label set carries two kinds of content, which we keep apart:

- **The Algebraic Skeleton:** Transition operators $T_\pm$ (radial) and $L_\pm$ (angular) generate the quantum numbers. The number operator $N = -2[T_+, T_-]$ and the Casimir L^2 built from them reproduce n exactly on the interior shells and $\ell(\ell+1)$ on every shell (on every shell, including the top one, the

$SO(4)$ Casimir n^2-1 is exact on the labels); the formula $E_n = -1/(2n^2)$ is then an imposed map from n through Fock's energy shell, with $p_0 = Z/n$ fixed by Fock's kernel eigenvalue (Paper 7), not an output of the ladder operators (once Fock's kernel fixes the function, it is $E_n = -1/(2(C+1))$ of the Casimir $C = n^2-1$ on the labels).

- **The Geometric Flesh:** The lattice *topology*—the pattern of connections—produces finite-truncation artifacts, not physical corrections: the structure of the graph Laplacian is intrinsic to its disconnected two-dimensional grid blocks. Node degrees differ between angular-momentum sectors at finite truncation — on the binary lattice they depend on m and on which shell neighbours exist, not monotonically on l (read before 2026-09-03 as differential weighted connectivity [retracted 2026-09-04: sp-splitting-aliasing-mechanism] lifting the s/p degeneracy; withdrawn — the l-blocks are disconnected components). On the optional topological-weight lattice the edge-weight log-holonomy measures the variation (a discrete gradient) of the weight function and decays as n^{-1} (it vanishes identically on the binary production lattice).

In this paper, we present computational evidence for this dual framework. Section II demonstrates exactness of the quantum numbers on interior shells. Section III analyzes the s/p node-amplitude proxy, showing it is a discretization artifact whose size tracks the residue of $n_{\max}$ modulo 3 — not a graph degeneracy (the ℓ-blocks are disconnected) and not a continuum-convergence signal. Section IV analyzes the geometric phase structure, showing that the Berry phase vanishes while the log-holonomy decays as n^{-1}. Section V discusses the separation between the algebraic and graph-geometric content. Section VI summarizes our findings.

II. THE ALGEBRAIC SKELETON: EXACT QUANTUM NUMBERS

The paraboloid lattice is generated by two sets of ladder operators forming the hydrogen atom's dynamical algebra $SU(2) \otimes SU(1,1)$:

$$\text{Angular:} \quad L_z, L_\pm \quad \text{(standard angular momentum)}, \tag{1}$$

$$\text{Radial:} \quad T_0, T_\pm \quad \text{(energy shell transitions)}. \tag{2}$$

These operators act on basis states $|n,l,m\rangle$ with selection rules:

$$L_\pm|n,l,m\rangle \propto |n,l,m\pm1\rangle, \tag{3}$$

$$T_\pm|n,l,m\rangle \propto |n\pm1,l,m\rangle. \tag{4}$$

The operator weights are the standard $SU(2)$ ladder matrix elements [5] and the noncompact $SU(1,1)$ ladder matrix elements [3].

A. Quantum-Number Reproduction

The critical test of this algebra is spectroscopic: do the operators reproduce the quantum numbers that label the Rydberg formula? **[SYMBOLIC PROOF]** They do, exactly, on interior shells: the operators return n ($n < n_{\max}$) and $\ell(\ell+1)$ (identity below), and the Rydberg energies

$$E_n = -\frac{1}{2n^2} \quad \text{(Hartree atomic units)}, \tag{5}$$

are then the imposed map from n given by Fock's energy shell $p_0^2 = -2E$, with $p_0 = Z/n$ fixed by Fock's kernel eigenvalue (Paper 7), not an eigenvalue output of the lattice. (The matched kinetic scale $\kappa = -1/16$ plays no role here: it matches only the graph Laplacian's extremal eigenvalue, a band edge, to $-\frac{1}{2}$, and its coincidence with the Fock Jacobian factor is an Observation, Paper 7.) No fitting parameters are required; the algebra reproduces the quantum-number labels (with n entering through the chosen $T_\pm$ weights).

This result validates the lattice as a faithful discrete *labelling* of the (truncated) hydrogen Hilbert space. The nodes are not a coarse-graining—they label the quantum states. The edges connect labels related by the ladder operators $T_\pm, L_\pm$; they are not dipole-allowed physical transitions. The quantum-number structure emerges from operator eigenvalues, not from solving differential equations; the energies are the imposed energy-shell map of n. (Verified in `tests/test_paper1_rydberg.py`: the $SU(1,1)$ number operator $N = -2[T_+, T_-]$, assembled purely from the ladder generators, has integer spectrum $\{n\}$ on the *interior* shells $n < n_{\max}$ and the $SU(2)$ Casimir L^2 gives $l(l+1)$, both exactly: with $w_n^2 = (n-l)(n+l+1)/4$ the squared T_+ weight, $[T_+, T_-]$ is diagonal with entry $w_{n-1}^2 - w_n^2 = -n/2$ on interior shells, so $N = n$, checked in exact arithmetic and to 10^{-9} in floating point; $E_n = -1/(2n^2)$ is then imposed as a function of n, not output by the algebra. This is the dynamical-symmetry construction, and it has three limits stated here so that it is not over-read: its n-dependence enters through the off-diagonal $T_\pm$ weights rather than through a diagonal operator; it is not exact on the top shell, where the truncated commutator makes $N = -(n-1-l)(n+l)/2$, l-dependent and non-positive (the test checks interior states only); and $SU(1,1)$ is noncompact, so its truncation is not a subrepresentation. The exact *compact* discrete operator on these labels is the $SO(4)$ Casimir $\mathbf{L}^2 + \mathbf{A}^2 = n^2 - 1$ ($\mathbf{A}$ the energy-normalized Runge–Lenz vector), diagonal with multiplicity n^2 on every shell including the top one, since each shell is a complete irreducible representation (`tests/test_graph_s3_nonconvergence.py`); it is $-\Delta_{S^3}$, the S^3 Laplace–Beltrami operator up to sign, acting on the label set. The discrete graph Laplacian $L = D - A$ is a separate object and does *not* reproduce this spectrum in any limit: it preserves l, splits into $n_{\max}$ disconnected blocks, and does not converge to Δ_{S^3}

(Paper 7 [10]; corrected 2026-09-25 from "reproduces the spectrum only in the continuum limit").)

III. THE GEOMETRIC FLESH: TOPOLOGICAL ARTIFACTS OF FINITE TRUNCATION

While the algebraic operators yield the exact quantum numbers (on interior shells), the *geometric structure* of the lattice—its connectivity pattern—introduces systematic artifacts when the lattice is truncated at finite $n_{\max}$. We demonstrate this by constructing the graph Laplacian and analyzing its finite-truncation properties as a function of lattice size.

A. The Graph Laplacian Hamiltonian

In graph theory, the kinetic energy of a particle hopping on a graph's vertices is described by the **graph Laplacian** [7]; here the vertices are quantum-number labels, not positions, so this is a statement about the graph, not about the electron:

$$L = D - A, \tag{6}$$

where A_{ij} is the adjacency matrix (edge weights) and D is the weighted degree matrix[11]. This Laplacian is a discrete analog of $-\nabla^2$ on each l-block (a two-dimensional grid in n and m); it is not a discretization of the S^3 Laplacian, to which it does not converge (Paper 7).

For the paraboloid lattice, we construct A from the transition operators:

$$A = |T_+| + |T_-| + |L_+| + |L_-|, \tag{7}$$

where $|\cdot|$ denotes element-wise absolute value (undirected edges). The Hamiltonian becomes

$$H = \beta L + V = \beta(D - A) + V, \tag{8}$$

where $V_{ii} = -1/n_i^2$ is a diagonal Coulomb node weight and β a scaling factor. **Eq.** (8) **is a general form, not the object measured below and not the production object** (clarified 2026-09-03): the splitting measurements of §III are taken on the bare graph Laplacian $L = D - A$, and the labelled legacy graph model `AtomicSolver(hamiltonian='graph')` builds $H = K_{\text{vac}}L$ with node weights never applied (the solver's default is now the physical diagonal $H = \text{diag}(-Z^2/2n^2)$ on the labels; the splitting numbers below are measured on the bare lattice and are unaffected). Under Eq. (8) with $V_{ii} = -1/n_i^2$ and $\beta = 1$, the first six of the twelve cutoffs of §III ($n_{\max}$ = 5–10) give 44.7, 2.2, 27.5, 10.6, 31.5 and 620.9% — a different quantity entirely (the values depend on β; $\beta = \pm 1/16$ gives different figures, and none of them is pinned by a test).

B. Differential Connectivity by Angular Momentum

The degree matrix D is *not rotationally invariant.* The weighted degrees originally printed for states with $n = 2$ were

$$\text{Pole (2s):} \quad D_{(2,0,0)} = 0.854, \tag{9}$$

$$\text{Equator (2p):} \quad D_{(2,1,0)} = 3.416, \tag{10}$$

historical values that reproduce under no current construction (flagged inline 2026-09-28; the binary production lattice gives $D_{2s} = 2$, $D_{2p} = 3$, the CG-magnitude rebuild 1.93 and 3.83 — see the §III QA note and Table I). **[MEASURED]** On the binary production lattice the degree is *not* monotone in l (nor is it on the CG construction of §III once $m \neq 0$ nodes are counted: $3d_{m=\pm2}$ 3.23 < $3p_{m=0}$ 5.41). A node's degree counts its $L_\pm$ neighbours (two for interior m, one at $|m| = l$, none for $l = 0$) plus its $T_\pm$ neighbours ($n \pm 1, l, m$) that exist; so $D_{2p,\,m=0} = 3 > D_{2s} = 2 = D_{2p,\,m=\pm1}$, while in an interior shell the maximum degree rises from $l = 0$, holds at its plateau value from $l = 1$ to $l = n - 2$, and drops at $l = n - 1$ ($n = 3$ at $n_{\max} = 4$: $l = 0, 1, 2 \to 2$, 3–4, 2–3; $n = 5$ at $n_{\max} = 6$: maxima $2, 4, 4, 4, 3$), and the top shell is lowered by truncation. A higher degree means a higher on-site entry in the Laplacian formulation; it does not couple the l-blocks, which are disconnected components, so it is not a mechanism for the s/p proxy (that reading withdrawn 2026-09-03; §III).

C. The Splitting as Discretization Artifact

We performed exact diagonalization of the bare graph Laplacian $L = D - A$ for lattices with $n_{\max}$ ranging from 5 to 30 (corrected 2026-09-03: this paragraph read "of H (Eq. 8)", which is not the operator measured — see the note there). Using a full dense eigendecomposition, we identified, for each of the nodes $|2, 0, 0\rangle$ (2s) and $|2, 1, 0\rangle$ (2p), the eigenvalue λ whose eigenspace projection $\|P_\lambda e_{\text{node}}\|$ is maximal (a basis-free rule: several eigenvalues are degenerate across l-blocks, so an argmax over the components of individual eigenvectors depends on the arbitrary basis chosen inside those eigenspaces — corrected 2026-09-28), and computed the relative splitting $\Delta E_{\text{rel}} = |E_{2p} - E_{2s}|/|E_{2s}|$.

At $n_{\max} = 10$ (385 nodes), the *originally reported* splitting was:

$$\lambda_{2s} = 0.0202, \tag{11}$$

$$\lambda_{2p} = 0.0238, \tag{12}$$

$$\Delta E = \lambda_{2p} - \lambda_{2s} = 0.0035, \tag{13}$$

quoted as a 16% relative splitting (arithmetically 17.3%). **[MEASURED]** *These are not the bare Laplacian's values* and are retained only as the historical record: on the operator named above, $L = D - A$ at $n_{\max} = 10$ gives

$\lambda_{2s} = 3.1756$, $\lambda_{2p} = 3.1206$ (corrected 2026-09-04 from a transposed 3.2306 — its mirror about λ_{2s}, which is why the ratio was unaffected) and a 1.73% ratio (see the QA correction below, and the twelve-cutoff list in §III). Flagged inline 2026-09-04: the method sentence above was corrected to name $L = D - A$ while this display still carried the legacy figures, so the corrected sentence appeared to vouch for them. However, this proxy gap is *not* an emergent physical force: the cutoff sweep below shows it is a **discretization artifact** of finite lattice truncation, with no spectral-aliasing mechanism (the ℓ-blocks are disconnected, so nothing is aliased; that analogy was retired 2026-09-03).

a. QA correction (2026-06-14). [**MEASURED**] The specific quantitative values in this section—the degrees $D_{2s} = 0.854$, $D_{2p} = 3.416$, the eigenvalues $\lambda_{2s} = 0.0202$, $\lambda_{2p} = 0.0238$, and the 16% splitting quoted at $n_{\max} = 10$—are *not* reproduced by the current production lattice (the $A = |T_+| + |T_-| + |L_+| + |L_-|$ CG-magnitude construction they assume is not implemented in `geovac/`; a faithful rebuild of that construction gives $D_{2s} \approx 1.9$ and $D_{2p,\, m=0} \approx 3.8$). Two lattices must be kept apart (2026-09-03): on the CG-weighted construction the $2s/2p$ splitting does *not* decay—129%, 68%, 84% and 139% at $n_{\max} = 8, 10, 15, 20$—whereas on the *binary* production lattice (`GeometricLattice`, all edge weights 1, the object used by the legacy graph mode `AtomicSolver(hamiltonian='graph')`) it is 37%, 12.7%, 0.65%, 15.6%, 5.1%, 1.7%, 2.7%, 1.7%, 1.1%, 4.1%, 1.8% and 0.39% at $n_{\max} = 5, 6, 7, 8, 9, 10, 12, 15, 18, 20, 25, 30$. Six of those twelve cutoffs are multiples of 3, including the endpoint, and on that residue class the value is exactly $\big(2 - 2\cos\frac{\pi}{n_{\max}-1}\big)/3$ (Eq. (14)); the neighbouring cutoffs 28 and 29 give 1.65% and 2.58%. What *is* verified (`tests/test_paper1_block_spectrum.py`, `test_trunk_qa_degree_table.py`; the older `test_trunk_qa_splitting.py` took a raw-eigenvector argmax, which is basis-dependent inside degenerate eigenspaces — flagged 2026-09-28 and rewritten the same day to the basis-free $\|P_\lambda e_{\rm node}\|^2$ rule, so it now counts as backing) is therefore: on the CG construction the $m = 0$ nodes satisfy $D_{2p} > D_{2s}$ and $D_{3p} > D_{3s}$ (on the binary lattice the degree is not monotone in l even at $m = 0$, e.g. $3p_{m=0}$ degree 4 against $3d_{m=0}$ degree 3 at $n_{\max} = 4$; on the CG construction it is not monotone once $m \neq 0$ nodes are counted), and on the binary lattice the s/p splitting decays, strongly oscillating, to below 1% by $n_{\max} = 30$. The specific CG-construction figures are retained as illustrative of that construction only. *This caveat applies wherever these figures recur—Table I (the abstract, Appendix B and the convergence list below now print the binary-lattice measurements). (The placeholder figure environments that also restated them were removed 2026-09-01.)*

Conditioning caveat (2026-09-03). The l-blocks of the graph Laplacian are *disconnected* components (no edge changes l), so there is no $2s/2p$ degeneracy in the graph spectrum to lift: the quantity reported here selects, for each of the nodes $|2,0,0\rangle$ and $|2,1,0\rangle$, the eigenspace of maximal projection at that node, and that selection is a near-tie (for the $2s$ node at $n_{\max} = 30$ the top two projection *amplitudes* $\|P_\lambda e_{\rm node}\|$ agree within 1.2%, i.e. within 2.4% as probabilities $\|P_\lambda e_{\rm node}\|^2$; the winner is $\lambda = 3.0$, and the runner-up is an exact three-way tie among $\lambda = 0.0110$, 2.8135 and 3.1756). The percentages below reproduce exactly but are *not robust* to the mode-selection rule; they are a node-amplitude proxy, not a spectral gap (`tests/test_paper1_block_spectrum.py`).

D. Convergence Analysis

The key diagnostic is the behavior of $\Delta E_{\rm rel}$ as $n_{\max} \to \infty$. [**MEASURED**] We swept the lattice cutoff from $n_{\max} = 5$ (55 vertices) through $n_{\max} = 30$ (9,455 vertices) and tracked the 2s/2p splitting at each resolution:

- At $n_{\max} = 5$: $\Delta E_{\rm rel} \approx 37\%$ (the maximum on the grid)
- At $n_{\max} = 6, 7$: 12.7%, 0.65%
- At $n_{\max} = 8$: 15.6% (a secondary peak); by $n_{\max} = 10$ it has fallen to $\approx 1.7\%$
- At $n_{\max} = 15, 18$: 1.7%, 1.1%
- At $n_{\max} = 20$: $\Delta E_{\rm rel} \approx 4.1\%$ (a third excursion)
- At $n_{\max} = 25, 30$: 1.8%, 0.39%

(binary production lattice, basis-free maximum-projection identification — see the conditioning caveat above; `tests/test_paper1_block_spectrum.py` pins the on-branch closed form, the 28/29/30 values, the conditioning and the CG-construction non-decay; the off-branch values at $n_{\max} = 5$–10, 20 and 25, and every value on this list, are now pinned by `tests/test_trunk_qa_splitting.py`, rewritten 2026-09-28 to the basis-free $\|P_\lambda e_{\rm node}\|^2$ rule (invariant under a rotation of the eigenbasis inside a degenerate eigenspace, unlike the raw-eigenvector argmax it replaces) and so counted as backing; the list printed before 2026-09-03—13%, 16%, 0.3%, 0.005%—did not reproduce at $n_{\max} = 5$, 20 or 30.)

The splitting does not decay monotonically; it *oscillates* at intermediate resolutions. [**INTERNAL THEOREM**] *The oscillation is the residue of $n_{\max}$ modulo 3, and the sequence is closed form* (2026-09-03). The $\ell = 0$ block is the path $P_{n_{\max}}$, whose eigenvalues are $2 - 2\cos(j\pi/n_{\max})$; the mode selected by maximal overlap with the $2s$ node has $j = \mathrm{round}(2n_{\max}/3)$, so

$$\lambda_{2s} = 3 \text{ exactly} \quad \Longleftrightarrow \quad n_{\max} \equiv 0 \pmod 3, \tag{14}$$

and on that branch the reported ratio is exactly $\big(2 - 2\cos\frac{\pi}{n_{\max}-1}\big)/3$ — 2.7005, 1.6715, 1.1351, 0.8208, 0.6209,

0.4861 and 0.3908% at $n_{\max} = 12, 15, 18, 21, 24, 27, 30$, matching the measured values to every digit printed. *At neighbouring cutoffs* the off-branch value is generically larger — asymptotically by $n_{\max}/(\sqrt{3}\,\pi)$ (the asymptotic itself stands; only two earlier *precision* claims about it were retracted [retracted 2026-09-04: sp-ratio-false-precision], see below), but only roughly at small cutoff: 4.2× and 6.6× at $n_{\max} = 28, 29$ against 30 (1.65%, 2.58% vs 0.39%), where the formula gives 5.5; 20–23× at 118, 119 against 120 (22.1); 550–552× at 2998, 2999 against 3000 (551.3). Below $n_{\max} \approx 12$ the ordering can invert (0.65% at 7 against 12.7% at 6). (Two earlier statements of this ratio — "three to six times" and "tracking $n_{\max}/(\sqrt{3}\,\pi)$ to within 1%" [retracted 2026-09-04: sp-ratio-false-precision] — were each read off three sampled cutoffs and stated as theorems; both are withdrawn.) **The endpoint quoted throughout this paper,** 0.39% **at** $n_{\max} = 30$**, therefore lies on the favourable branch**, a selection this paper did not disclose before 2026-09-03.

The decay is consequently $O(n_{\max}^{-2})$ along $n_{\max} \equiv 0$ and $O(n_{\max}^{-1})$ generically — a proved rate, replacing the earlier statement that none was proven. *Proof sketch of the generic rate:* off the branch $|j - 2n_{\max}/3| = 1/3$, so $\lambda_{2s} - 3 = \pm 2\sin(2\pi/3)\,\pi/(3n_{\max}) + O(n_{\max}^{-2}) = \pm\pi/(\sqrt{3}\,n_{\max}) + O(n_{\max}^{-2})$, which dominates the $O(n_{\max}^{-2})$ $2p$-side term; dividing by the on-branch value gives the $n_{\max}/(\sqrt{3}\,\pi)$ ratio. (The on-branch closed form is pinned by a test; this generic $O(n_{\max}^{-1})$ rate currently has no dedicated test.) *The mechanism previously printed here — "spectral aliasing… standing-wave reflections that differentially [retracted 2026-09-04: sp-splitting-aliasing-mechanism] shift eigenvalues in distinct angular momentum sectors" — is withdrawn:* the ℓ-blocks are disconnected components (`tests/test_paper1_block_spectrum.py`), so there is no s/p degeneracy present to alias and no cross-sector reflection to shift it. Both selected eigenvalues approach $3 = 2 - 2\cos(2\pi/3)$ because the target node is vertex 2 of a path in each block. As $n_{\max}$ grows the two proxy eigenvalues coincide (a path-graph coincidence), which does *not* reproduce the continuum $SO(4)$ Coulomb degeneracy (the ℓ-blocks are disconnected, so the graph has no s/p degeneracy of its own).

E. Physical Interpretation

[MEASURED] The 2p, $m = 0$ node has the higher degree ($D_{2p,\,m=0} = 3 > D_{2s} = 2$; the $m = \pm 1$ nodes tie with $2s$; the "4×" ratio came from the non-reproducing Table I figures — see the §III QA note). *Read before 2026-09-03 as:* the higher-degree 2p node is more sensitive to the lattice boundary than the 2s state, the finite-$n_{\max}$ truncation acting as an absorbing wall that differentially perturbs high-connectivity nodes [retracted 2026-09-04: sp-splitting-aliasing-mechanism]. *Withdrawn for the s/p proxy (2026-09-03):* its size is set by the residue of $n_{\max}$ modulo 3, not by a boundary effect; the "geometric boundary effect" reading printed here until 2026-09-04 is withdrawn with it, and it is not an emergent centrifugal barrier either.

Crucially, *no explicit* $l(l+1)/r^2$ term appears in the Hamiltonian. The transient splitting is a finite-truncation effect of the node-amplitude proxy (its size set by the residue of $n_{\max}$ modulo 3), not a force encoded in the topology. The graph Laplacian does not spontaneously generate the Lamb shift or replace electromagnetic interactions, and its finite-lattice s/p proxy does *not* reproduce the continuum $SO(4)$ Coulomb degeneracy (the ℓ-blocks are disconnected components, so the graph Laplacian has no continuum limit carrying that degeneracy—it does not converge to Δ_{S^3}, Paper 7; the degeneracy is carried exactly by the $SO(4)$ Casimir on the labels, §II). **[INTERNAL THEOREM]** The s/p node-amplitude proxy is a characterisable discretization artifact with a closed-form value on $n_{\max} \equiv 0 \pmod 3$ and a proved rate — $O(n_{\max}^{-2})$ there, $O(n_{\max}^{-1})$ generically (Eq. (14); the earlier "no convergence rate is proven" is superseded). Its magnitude depends on the residue of $n_{\max}$ modulo 3 rather than on proximity to a continuum limit, so it does not function as evidence of convergence; see the §III QA note.

IV. GEOMETRIC PHASE STRUCTURE OF THE LATTICE

The lattice carries a natural notion of parallel transport around closed loops (plaquettes). We examine whether this transport produces a nontrivial geometric phase—a Berry holonomy [6]—and find that the answer is negative for the real-valued transition operators used here. We then characterize the *log-holonomy*, which measures variation (a discrete gradient) of the edge-weight function, and discuss what would be required for a genuine geometric phase.

A. Flat Connection for Real Operators

A **rectangular plaquette** on the lattice is constructed via radial ($T_\pm$) and azimuthal ($L_\pm$) operators:

$$|n,l,m\rangle \xrightarrow{T_+} |n{+}1,l,m\rangle \xrightarrow{L_+} |n{+}1,l,m{+}1\rangle \xrightarrow{T_-} |n,l,m{+}1\rangle \xrightarrow{L_-} |n,l, \tag{15}$$

The Berry phase for this plaquette is the argument of the holonomy product:

$$\theta = \arg\big(\langle n{+}1,l,m|T_+|n,l,m\rangle\,\langle n{+}1,l,m{+}1|L_+|n{+}1,l,m\rangle\,\langle n,l,m{+}1|T_-| \tag{16}$$

Each matrix element is a Biedenharn–Louck coupling coefficient. For the $SU(2)$ angular-momentum operators, $\langle l,m+1|L_+|l,m\rangle = \sqrt{(l-m)(l+m+1)}$, and for the $SU(1,1)$ radial operators, $\langle n+1,l|T_+|n,l\rangle =$

$\sqrt{(n-l)(n+l+1)/4}$. **[SYMBOLIC PROOF]** All four factors are **real and positive**. Therefore

$$\theta = \arg(\text{positive real number}) = 0 \qquad \text{for every plaquette at every } (n,l,m) \tag{17}$$

The lattice connection defined by these operators is *flat*: it carries zero Berry curvature.

Erratum. An earlier version of this paper reported a fitted exponent $k = 2.113 \pm 0.015$ for the Berry phase scaling. That value was a theoretical prediction placed as a placeholder pending numerical validation; the accompanying figure was never generated (it remained a `[Placeholder]`). Subsequent computation confirms $\theta = 0$ identically for all plaquettes, as shown above. The claimed $k = 2.113$ is therefore **retracted**.

B. Log-Holonomy of Edge Weights

Although the Berry phase vanishes, the lattice edge weights carry a well-defined *log-holonomy*—the logarithmic circulation around a plaquette:

$$\Theta(n) = \ln w_{12} + \ln w_{23} - \ln w_{34} - \ln w_{41}, \tag{18}$$

where w_{ij} are the edge weights of the *topological-weight* lattice (`GeometricLattice(topological_weights=True)`); on the binary production lattice and on the CG-magnitude construction of §III the T-edge weight is m-independent and the L-edge weight is n-independent, so $\Theta \equiv 0$ there (2026-09-03). **[SYMBOLIC PROOF]** For the topological weights $w = 1/(n_1 n_2)$, the $T_\pm$ edges share the factor $1/\big(n(n+1)\big)$ and cancel, leaving

$$\Theta(n) = \ln\frac{1}{(n+1)^2} - \ln\frac{1}{n^2} = -2\ln\frac{n+1}{n} \;\sim\; -\frac{2}{n} \qquad (n \gg 1). \tag{19}$$

Both results—$\theta = 0$ on every plaquette (with a phased-plaquette control confirming the detector is not vacuous) and the exact $\Theta(n)$ closed form with its n^{-1} decay—are pinned by `tests/test_paper1_geometric_phase.py` (added 2026-09-01, closing a coverage gap: the prior Berry-phase artifact was archived with the retracted $k = 2.113$ claim). The per-plaquette log-holonomy $\Theta(n) = -2\ln((n+1)/n)$ decays as n^{-1} asymptotically (a closed form, so no fitted exponent or R^2 is involved). This measures the **variation of the weight function** (a first difference of its logarithm) —how edge weights vary across the lattice—not a genuine geometric phase in the Berry sense.

C. Toward a Genuine Geometric Phase

A nonzero Berry phase requires complex-valued transition amplitudes. Possible extensions include:

1. *Not* a change of phase convention. Departing from the Condon–Shortley convention (which makes the $L_\pm$ matrix elements real and non-negative, as used above), e.g. by omitting its $(-1)^m$ factor on $m \geq 0$, flips the sign of every $L_\pm$ matrix element between two $m \geq 0$ states (of every $L_\pm$ element, if the factor is dropped for all m); but such a change is a diagonal ± 1 rephasing of the basis, and every closed loop visits each of its nodes twice (in and out), so every plaquette holonomy is unchanged and stays at $\theta = 0$. A convention change cannot produce a geometric phase (*corrected 2026-09-28;* the item previously claimed discrete $\{0, \pi\}$ phases). Genuine routes are:

2. A $U(1)$ gauge extension—attaching phases $e^{i\theta_e}$ to the *edges* (the transition operators) with nonzero net flux $\sum_{e\in\partial P}\theta_e \neq 0$ through some plaquette P—which would produce continuously varying curvature (phases that depend only on the node labels, $e^{if(n,l,m)}$ at each endpoint, are a rephasing of the basis and give zero holonomy, as in the item above);

3. An adiabatic Berry connection from eigenstates of a parameter-dependent Hamiltonian (as in the hyperspherical $P_{\mu\nu}(R)$ connection of Ref. [9]).

[OPEN] Whether any such extension recovers a scaling exponent near $k \approx 2$ (the velocity-dependent kinematic factor $v^2 \propto n^{-2}$) remains an open question for future work.

V. DISCUSSION: ALGEBRAIC VERSUS GRAPH-GEOMETRIC CONTENT

Our results separate two kinds of content carried by the label lattice.

A. The Two Kinds of Content

1. **Algebraic structure:** The operators $T_\pm, L_\pm$ and the number operator and Casimir built from them reproduce the *unperturbed* hydrogen quantum-number structure exactly on interior shells (and the $SO(4)$ Casimir $n^2 - 1$ is exact on every shell); the energies (Eq. 5) are the imposed energy-shell map of n, not an output of the ladder operators (once Fock's kernel fixes the function, they are $E_n = -1/(2(C+1))$ of the Casimir $C = n^2 - 1$ on the labels). This is the exact aspect, and it acts on the discrete labels.

2. **Graph-geometric structure:** The graph Laplacian $L = D - A$ carries finite-truncation structure that is intrinsic to its disconnected two-dimensional grid blocks; it supplies no physical correction or perturbation to the hydrogen levels. The weighted degree matrix D differs between angular momentum sectors at finite trunca-

tion (read before 2026-09-03 as differential connectivity [retracted 2026-09-04: sp-splitting-aliasing-mechanism] lifting the s/p degeneracy; withdrawn — the sectors are disconnected components). On the optional topological-weight lattice the edge-weight log-holonomy decays as n^{-1} (Section IV), measuring the variation (a discrete gradient) of that weight function; on the binary production lattice it vanishes.

The separation is the point: the algebraic operators provide the exact quantum-number structure, while the graph topology introduces finite-size artifacts that must be controlled.

B. Why Calibration Fails

When we attempt to match the graph Laplacian's eigenvalues directly to physical energies, calibration fails. The eigenvalues are non-negative (Section III), but that alone is not the obstruction — a negative scale such as $\kappa = -1/16$ fixes the sign and matches the extremal eigenvalue to $-\frac{1}{2}$.

The obstruction is structural: the Laplacian's spectrum is the band structure of its disconnected two-dimensional grid blocks, lying in $[0, 8]$, and it carries neither absolute energies nor physical splittings between hydrogen levels (Paper 7). The algebraic operators provide the quantum numbers, and the energies are the imposed energy-shell map of n; the graph Laplacian supplies no correction to them.

C. The Absence of Fine Structure

[MEASURED] A computational search (three checks) for the fine structure constant $\alpha \approx 1/137$ within the lattice geometry yields a null result. We examined:

1. **Operator ratios: [PANEL-VERIFIED]** The spectral-norm gearing ratio $\|L_+\|/\|T_+\| = 2.0$ *exactly*—at every lattice cutoff tested (and for every $N \geq 2$, since L_+ and T_+ are weighted shifts whose largest weights are $\sqrt{(N-1)N}$ and $\sqrt{(N-1)N}/2$; this argument is not separately test-backed)—while the Frobenius-norm ratio converges to $2\sqrt{6}/3 = 1.632993\ldots$, its exact limit (algebraic closed form; the frozen `test_trunk_qa_gearing.py` panel verifies only the monotone approach to $n_{\max} = 30$, ratio 1.701, not the limit) from $\|L_+\|_F^2/\|T_+\|_F^2 = 4(N+2)(2N+1)/(3N^2-2) \to 8/3$ (corrected 2026-09-03 from "drifts through ≈ 1.77 without converging": the sequence is monotone decreasing and 1.77 is a finite-$n_{\max}$ value, not a non-limit) (`tests/test_trunk_qa_gearing.py`). Neither encodes a physical constant such as 137 or $1/\alpha$: the spectral value is 2, and the Frobenius limit is the algebraic number $2\sqrt{6}/3$. The commutator $[T_+, L_+] = 0$, however, *is* bit-exact.

2. **Commutator defects: [PANEL-VERIFIED]** The operators commute exactly, $[T_+, L_+] = 0$ (the radial and angular ladders act on independent labels), leaving no commutator defect in which a coupling constant could sit.

3. **Holonomy defects: [SYMBOLIC PROOF]** The Berry phase is identically zero for real operators (Section IV); on the topological-weight lattice the log-holonomy decays as n^{-1}, reflecting the asymptotic flatness of that weight function.

This null result is *physically correct*. The fine structure constant $\alpha = e^2/(4\pi\epsilon_0\hbar c)$ measures the strength of electromagnetic interactions—specifically, the coupling between electron and photon fields. Our lattice describes the *kinematic* structure of the electron state space without electromagnetic interactions. The hydrogen spectrum $E_n = -1/(2n^2)$ is independent of α; fine structure appears only through:

$$\text{Relativistic corrections:} \quad \Delta E_{\text{rel}} \sim \alpha^2 \cdot E_n/n, \tag{20}$$

$$\text{Spin-orbit coupling:} \quad \Delta E_{\text{so}} \sim \alpha^2 \cdot E_n \cdot l(l+1)/n^3, \tag{21}$$

$$\text{Lamb shift (QED):} \quad \Delta E_{\text{Lamb}} \sim \alpha^5 \cdot mc^2. \tag{22}$$

These require photon degrees of freedom, which are absent from our single-particle lattice.

This clarifies the role of geometry: the lattice encodes *kinematic* constraints (quantum-number labels, angular momentum structure) but not *dynamical* couplings. **[OPEN]** (Speculation, untested:) fine structure would require additional photon degrees of freedom coupled to the electron labels, and in such a construction α would enter as a coupling strength supplied from outside the electron lattice; no such construction is built here, and the one attempt to derive α from a lattice ratio is recorded in §VII as superseded.

D. Artifact vs. Physics: Drawing the Line

The cutoff sweep separates discretization artifacts from the structural (non-physical) content of the graph Laplacian, without being evidence of convergence to any continuum operator. The s/p splitting is firmly in the artifact category: on the binary lattice it decays as $n_{\max} \to \infty$ (not on the CG-weighted construction of §III), oscillating with the residue of $n_{\max}$ modulo 3 (no aliasing mechanism is involved; that reading was withdrawn 2026-09-03).

By contrast, the spectral bound $\lambda_{\max} \to 2d_{\max} = 8$ (hence the extremal eigenvalue $\kappa\lambda_{\max} \to -1/2$; the graph spectrum is *not* a level-by-level Rydberg ladder,

`tests/test_paper7_graph_convergence.py`) is a *convergent* property of the graph as $n_{\max} \to \infty$; the log-holonomy $\Theta(n)$ of the topological-weight lattice (Section IV) is a closed-form function of the shell index n, not of $n_{\max}$, and decays as n^{-1}. They are structural content of the graph itself, not of S^3: the bound $8 = 2d_{\max}$ is the band edge of any large two-dimensional grid, and the graph Laplacian does not converge to Δ_{S^3} (Paper 7; the "survive the continuum limit" reading printed here until 2026-09-25 is withdrawn).

This distinction is essential: the graph Laplacian does not generate electromagnetic forces or the Lamb shift, and it does not carry the *kinematic* structure of the electron state space either. That structure lives in the label set (n, ℓ, m) and its ladder operators $T_\pm, L_\pm$ (with the $SO(4)$ Casimir $n^2 - 1$ diagonal on the labels); the graph Laplacian carries no S^3 or hydrogen spectral content, only the structure of its disconnected two-dimensional grid blocks. Finite lattice effects must be understood as numerical artifacts, not new physics.

E. The Wavefunction as Distribution

If quantum states are labelled by discrete nodes, what is the wavefunction $\psi(r)$? In this framework it is the standard object: a probability amplitude expanded in the basis states the nodes label, each node carrying one coefficient. A reading in which ψ is a statistical distribution over lattice sites and the Schrödinger equation becomes a diffusion equation on the graph is *not* available here: the graph Laplacian does not converge to Δ_{S^3} and its spectrum does not carry the Rydberg ladder (Paper 7; §III), so diffusion on the graph is not hydrogen dynamics (reading withdrawn 2026-09-28).

VI. CONCLUSION

We have shown that the hydrogen state space, when labelled on a discrete lattice, carries two separable kinds of content:

- **Algebraic exactness: [SYMBOLIC PROOF]** Transition operators reproduce the hydrogen quantum numbers exactly on interior shells ($n < n_{\max}$; a commutator identity, checked in exact arithmetic) and $\ell(\ell+1)$ on every shell; the $SO(4)$ Casimir n^2-1 is exact on every shell; the energies are the imposed energy-shell map of n, with $p_0 = Z/n$ fixed by Fock's kernel eigenvalue (Paper 7), not an output of the ladder operators (once Fock's kernel fixes the function, it is $E_n = -1/(2(C+1))$ of the Casimir $C = n^2 - 1$ on the labels).

- **Understood discretization artifacts: [INTERNAL THEOREM]** The graph Laplacian's node-amplitude proxy for the s/p gap is closed form on $n_{\max} \equiv 0 \pmod 3$ and is set by that residue rather than by proximity to a continuum limit (2026-09-03). **[MEASURED]** It is large at finite $n_{\max}$ (up to 37% at $n_{\max} = 5$ on the binary production lattice, 15.6% at 8, 1.7% at 10), and is not spectral aliasing [retracted 2026-09-04: sp-splitting-aliasing-mechanism] (withdrawn 2026-09-03: the l-blocks are disconnected and carry no degeneracy to alias) nor an emergent physical force. The proxy oscillates (4.1% at $n_{\max} = 20$) and reads 0.39% at $n_{\max} = 30$ on the $n_{\max} \equiv 0 \pmod 3$ branch (1.65%, 2.58% at 28, 29) — a proxy value, not a recovery of the exact Coulomb degeneracy [retracted 2026-09-04: sp-splitting-aliasing-mechanism], which belongs to the continuum operator; on the CG-weighted construction of §III the lift does *not* decay (129, 68, 84 and 139% at $n_{\max} = 8, 10, 15, 20$; `tests/test_paper1_block_spectrum.py`), so the artifact-decay claim is scoped to the binary lattice, and both figures carry the conditioning caveat of §III.

- **Geometric phase structure: [SYMBOLIC PROOF]** The Berry phase vanishes identically for the real-valued $SU(2)/SU(1,1)$ operators; on the optional topological-weight lattice the edge-weight log-holonomy decays as $\Theta \propto n^{-1}$, measuring the variation (a discrete gradient) of that weight function (it vanishes identically on the binary production lattice). Whether a $U(1)$ gauge extension can produce a genuine geometric phase with relativistic scaling remains an open question.

- **Kinematic completeness: [OBSERVATION]** The label set and its ladder operators encode the α-independent single-electron quantum-number structure—the Coulomb quantum numbers and the angular momentum structure (the Berry phase being zero)—while cleanly separating finite-truncation artifacts from physical content.

These results support a discrete label-set reformulation of the hydrogen state space in which the algebraic operators carry the exact quantum-number structure on interior shells (the energies being the imposed energy-shell map of n) and the graph Laplacian introduces controlled, quantitatively understood finite-size artifacts. Critically, the graph Laplacian does not replace electromagnetic interactions or generate the Lamb shift; the kinematic structure of the electron lives in the label set and its ladder operators, while the graph Laplacian carries no S^3 or hydrogen spectral content.

VII. LIMITATIONS AND NEXT STEPS: MISSING PHOTON DEGREES OF FREEDOM

What This Model Does Not Include:

The paraboloid lattice presented here is explicitly a *single-particle* model describing only the electron's quantum state manifold. It does not include:

1. **Photon degrees of freedom:** The electromagnetic field ($U(1)$ gauge symmetry) is absent.

2. **Electron-photon coupling:** There is no mechanism for radiative transitions or electromagnetic interactions.

3. **The fine structure constant α:** As a pure electron-only model, α does not appear in it (the null search of §V); α measures the strength of electron-photon coupling, which this model does not contain.

What would have to be added:

Incorporating electromagnetic interactions would require photon degrees of freedom and electron–photon coupling rules on top of the label lattice; no such construction is built or tested in this paper.

An earlier route, superseded. An earlier version of this section conjectured that α emerges as a "symplectic impedance" — a ratio of action densities between the electron paraboloid and a helical photon fiber. That route was tried: on the paraboloid lattice it reached 6.2×10^{-5} relative error only with a *fitted* helical pitch, and it has been superseded (Paper 2 [8] records the attempt and its supersession); it is not an open, unexplored route. **[OBSERVATION]** The companion paper [8] records a different, parameter-free numerical observation: α approximately satisfies (to 8.8×10^{-8} relative; an Observation, not a law) the cubic $\alpha^3 - K\alpha + 1 = 0$ with $K = \pi(B + F - \Delta)$ built from three spectral quantities on S^3. The combination rule is an Observation only, with no first-principles derivation; the companion paper's working hypothesis reads α as a projection constant, not derivable from any single principle.

Outlook:

This framework invites generalization. The hydrogen labels are those of $SO(4)$; what label sets describe molecules, nuclei, or quantum fields? The answer may lie in higher-dimensional lattices—label sets of larger symmetry groups.

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Appendix A: Mathematical Definitions

1. Weighted Degree Matrix

The weighted degree matrix D is a diagonal matrix where each diagonal element represents the total connection strength of a node:

$$D_{ii} = \sum_j |A_{ij}|, \tag{A1}$$

where A_{ij} is the adjacency matrix constructed from transition operators. For complex-valued operators, the absolute value ensures symmetric (undirected) edge weights. On the CG-magnitude construction the degrees are therefore non-integer (weighted sums of transition matrix elements); on the binary production lattice every edge weight is 1 and the degrees are integer edge counts (Table I caption).

2. Berry Phase and Log-Holonomy

For a plaquette (Eq. 15), the Berry phase is $\theta = \arg(\prod_i \langle s_i|\hat{U}_i|s_{i+1}\rangle)$. As shown in Section IV, all four matrix elements are real and positive, so $\theta = 0$ identically.

The log-holonomy (Eq. 18) provides a nonzero measure of weight-function variation. For topological weights $w = 1/(n_1 n_2)$, the exact result is $\Theta(n) = -2\ln\big((n+1)/n\big)$, decaying as n^{-1}.

Appendix B: Computational Details

1. Lattice Construction

The lattice was constructed for $1 \le n \le n_{\max}$ with $n_{\max}$ from 5 to 30 (at $n_{\max} = 10$, 385 quantum states $|n, l, m\rangle$ with $0 \le l < n$ and $-l \le m \le l$). Transition operators $T_\pm$ and $L_\pm$ were built from the standard $SU(2)$ and $SU(1,1)$ ladder matrix elements.

2. Graph Laplacian Diagonalization

The adjacency matrix A was computed as the sum of absolute values of all transition operators:

$$A = |T_+| + |T_-| + |L_+| + |L_-|. \tag{B1}$$

This CG-magnitude construction is used only for the §III comparison with the weighted lattice; its weighted degree matrix D ranges from 0.71 to 19.9 with mean degree 12.0 at $n_{\max} = 10$ (faithful rebuild, 2026-09-03; the figures printed earlier, 0.854–20.624 and 12.42, belong to the non-reproducing table of §III). The s/p proxy measurements of §III use the *binary* production lattice (every edge weight 1) and a full dense eigendecomposition of

$L = D - A$, matching §III (corrected 2026-09-28: this appendix previously described a shift-invert Lanczos computation of the lowest 20 eigenvalues on the CG-weighted lattice, which is not how the §III numbers were obtained).

For the scaling measurement only, a sparse CSR build with the shift-invert Lanczos method (`scipy.sparse.linalg.eigsh`, $\sigma = -1.0$, lowest 20 eigenvalues) was used; **[MEASURED]** the sparse-eigenvalue timing is guarded sub-quadratic and the adjacency nonzero count near-linear (measured exponent 1.07 over $n_{\max} = 5$–30) by `tests/test_ov_scaling_rigorous.py` (multi-point log–log fits; the frozen bounds are < 1.8 on timing and < 1.15 on the nonzero count).

State identification was performed by computing, for each distinct eigenvalue λ of the full spectrum, the eigenspace projection $\|P_\lambda |n, l, m\rangle\|^2$ of each target node, and selecting the eigenvalue with maximal projection (basis-free; an argmax over the components of individual eigenvectors is not well defined when eigenvalues are degenerate across l-blocks — corrected 2026-09-28; pinned by `tests/test_trunk_qa_splitting.py`, which checks invariance under a rotation of the eigenbasis inside a degenerate eigenspace on the fast grid, $n_{\max} = 6$–18; at $n_{\max} = 20, 25, 30$ it checks instead that the eigenvalue clustering is cleanly separated from numerical noise and that the printed values are stable across clustering tolerances).

3. Log-Holonomy Calculation

Rectangular plaquettes (Eq. 15) were identified by searching for closed paths in (n, m) space at fixed l. The Berry phase $\theta = \arg(\cdot)$ is identically zero for all plaquettes because every $SU(2)/SU(1,1)$ matrix element is real and positive (Section IV). The log-holonomy $\Theta(n) = -2\ln\big((n+1)/n\big)$ was computed analytically and verified numerically; its closed form decays as n^{-1} asymptotically (no fitted exponent is involved—a finite-n log–log fit gives -0.99 on $n = 10$–320).

Appendix C: Supplementary Data

[1] I. A. Malkin and V. I. Man'ko, "Symmetry of the hydrogen atom," JETP Lett. **2**, 146 (1965).

[2] C. Fronsdal, "Infinite multiplets and the hydrogen atom," Phys. Rev. **156**, 1665 (1967).

[3] A. O. Barut and H. Kleinert, "Transition probabilities of the hydrogen atom from noncompact dynamical groups," Phys. Rev. **156**, 1541 (1967).

[4] V. Fock, "Zur Theorie des Wasserstoffatoms," Z. Phys. **98**, 145 (1935).

[5] L. C. Biedenharn and J. D. Louck, *Angular Momentum in Quantum Physics*, Encyclopedia of Mathematics and its Applications, Vol. 8 (Addison-Wesley, Reading, MA, 1981).

[6] M. V. Berry, "Quantal phase factors accompanying adiabatic changes," Proc. R. Soc. Lond. A **392**, 45 (1984).

[7] F. R. K. Chung, *Spectral Graph Theory*, CBMS Regional Conference Series in Mathematics, Vol. 92 (American Mathematical Society, Providence, RI, 1997).

[8] J. Loutey, "The Fine Structure Constant as a Spectral Coincidence on the Hopf Fibration," (companion paper, 2026).

[9] J. Loutey, "The Hyperspherical Lattice: Two-Electron Atoms as Coupled Channel Graphs," (companion paper, 2026).

[10] J. Loutey, "The Dimensionless Vacuum: Recovering the Schrödinger Equation on a Scale-Invariant Label Lattice," GeoVac Paper 7 (2026).

[11] The weighted degree $D_{ii} = \sum_j |A_{ij}|$ represents the total connectivity strength of node i. For transition matrices with complex coefficients, we use absolute values to obtain undirected edge weights, explaining the non-integer values in Table I.

TABLE I. **Historical values — not reproduced by any current construction** (flagged 2026-09-03). These figures are retained to document the original calculation; the production binary lattice gives *integer* degrees — $D_{2s} = 2$ and $D_{2p} = 3$ exactly, every edge weight being 1 — while the CG construction gives 0.7071, 1.9319, 3.8284, 2.9568, 5.4096, 6.1237, of which only the $3d$ row here reproduces. (Corrected 2026-09-04: this caption assigned the CG values 1.9 and 3.8 to the binary lattice, inverting the separation it was written to enforce.) What survives is the *qualitative* ordering $D_{2p,\,m=0} > D_{2s}$, which both constructions share (on the binary lattice the $m = \pm 1$ nodes tie with $2s$). Weighted node degrees for select quantum states. The non-integer values arise from weighted sums of transition matrix elements (see Appendix A).

State (n, l, m)	Weighted Degree D_{ii}	Type	Laplacian eigenvalue (historical)
$(1, 0, 0)$	0.854	1s	(ground state)
$(2, 0, 0)$	0.854	2s	0.0202
$(2, 1, 0)$	3.416	2p	0.0238
$(3, 0, 0)$	0.854	3s	—
$(3, 1, 0)$	4.270	3p	—
$(3, 2, 0)$	6.124	3d	—